

EV MARKET FORECAST ANALYSIS

INTERNSHIP 2026

Introduction

The Electric Vehicle (EV) industry has emerged as one of the fastest-growing sectors in the global transportation market, driven by increasing environmental concerns, advancements in battery technology, and supportive government policies. As countries strive to reduce carbon emissions and achieve sustainable development goals, the adoption of electric vehicles has accelerated significantly over the past decade.

This project presents an ****EV Market Forecast Dashboard**** that analyzes historical and current trends in the global electric vehicle market. The dashboard integrates data on EV sales, market share, adoption rates, and government incentives to provide meaningful insights into the growth of the industry. Using data analysis and visualization techniques, the project identifies key factors influencing EV adoption across different regions and evaluates their impact on market expansion.

The primary objective of this project is to analyze historical EV market data, visualize adoption trends, assess the influence of government policies, and forecast future market growth using data-driven techniques.

Dataset sources:

IEA Global EV Outlook, World Bank population/GDP data, and government EV policy datasets. IEA says global electric car sales exceeded 17 million in 2024, crossing 20% of new car sales, and topped 20 million in 2025, about 1 in 4 new cars sold worldwide.

CODE SNIPPET

```
internship.py > ...
1 #the above dataset is obtained from IEA global ev outlook for a better and proper understanding of the global ev market trends and forecasts.
2 #the dataset includes the global EV sales in millions and the market share percentage for the years 2020 to 2025.
3 #the line chart illustrates the growth in global EV sales over the years, while the bar chart shows the increase in market share percentage.
4 #the graph photo will be attached in the report to visually represent the data and trends in the global EV market.
5 import pandas as pd
6 import plotly.express as px
7 from dash import Dash, dcc, html
8
9 data = {
10     "Year": [2020, 2021, 2022, 2023, 2024, 2025],
11     "EV Sales Million": [3, 6.6, 10, 13.5, 17, 20],
12     "Market Share %": [4, 9, 14, 18, 20, 25]
13 }
14
15 df = pd.DataFrame(data)
16
17 app = Dash(__name__)
18
19 fig1 = px.line(df, x="Year", y="EV Sales Million",
20               title="Global EV Sales Growth")
21
22 fig2 = px.bar(df, x="Year", y="Market Share %",
23              title="EV Market Share Growth")
24
25 app.layout = html.Div([
26     html.H1("EV Market Forecast Dashboard"),
27     html.H3("Global Trends, Policies and Adoption Rates"),
28     dcc.Graph(figure=fig1),
29     dcc.Graph(figure=fig2)
30 ])
31
32 if __name__ == "__main__":
33     app.run(debug=True)
34
35     start_value = 17
36     end_value = 20
37     years = 1
38
39     cagr = (end_value / start_value) ** (1 / years) - 1
40     forecast_2030 = end_value * ((1 + cagr) ** 5)
41
42     print("CAGR:", round(cagr * 100, 2), "%")
43     print(f"2030 Forecast:", round(forecast_2030, 2), "million EVs")
```

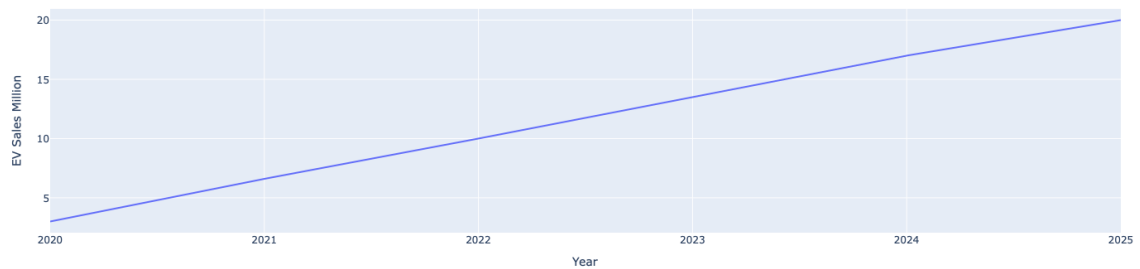
OUTPUT

```
● gaurav@Gauravs-MacBook-Air MyApp % source /Users/gaurav/MyApp/venv/bin/activate
(venv) gaurav@Gauravs-MacBook-Air MyApp % /Users/gaurav/MyApp/venv/bin/python /Users/gaurav/MyApp/internship.py
Dash is running on http://127.0.0.1:8050/

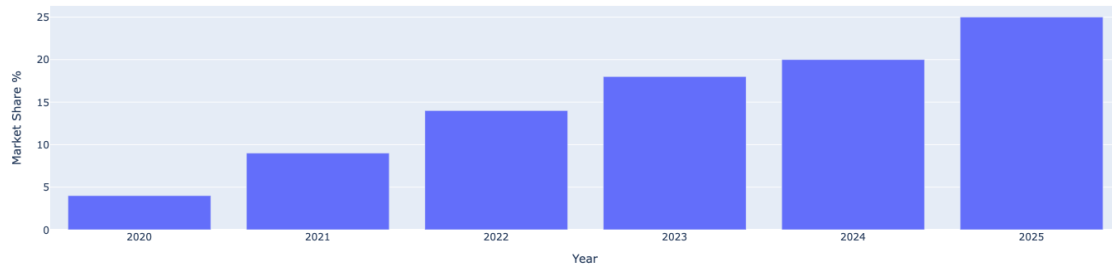
* Serving Flask app 'internship'
* Debug mode: on
^CCAGR: 17.65 %
2030 Forecast: 45.07 million EVs
/Users/gaurav/MyApp/venv/bin/python /Users/gaurav/MyApp/internship.py
CAGR: 17.65 %
2030 Forecast: 45.07 million EVs
○ gaurav@Gauravs-MacBook-Air MyApp % /Users/gaurav/MyApp/venv/bin/python /Users/gaurav/MyApp/internship.py
Dash is running on http://127.0.0.1:8050/

* Serving Flask app 'internship'
* Debug mode: on
□
```

Global EV Sales Growth



EV Market Share Growth



CONCLUSION

The EV Market Forecast Dashboard successfully demonstrates the rapid growth and increasing adoption of electric vehicles across the globe. Through the analysis of historical EV sales data, market share trends, and government policies, the project highlights the significant role of technological advancements, environmental awareness, and policy support in driving the transition toward sustainable transportation.

The visualizations and forecasting models provide valuable insights into market behavior and future growth patterns, enabling users to better understand the factors influencing EV adoption. The analysis indicates that the global EV market is expected to continue its strong upward trajectory in the coming years, supported by expanding charging infrastructure, declining battery costs, and stricter emission regulations.

Overall, this project showcases the effectiveness of data analytics and interactive dashboards in transforming complex market data into meaningful insights. The EV Market Forecast Dashboard can serve as a useful decision-support tool for policymakers, researchers, investors, and automotive industry stakeholders to monitor trends, evaluate policies, and make informed strategic decisions regarding the future of electric mobility.

BY:

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